

Optimal bidding strategy of renewable-based virtual power plant in the day-ahead market

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ARTICLE INFO

Keywords:

Virtual power plant
Uncertainty modeling
Bidding strategy
Energy market
Ancillary service markets

ABSTRACT

This paper proposes an optimal bidding strategy model of a virtual power plant (VPP) in the day-ahead market (DAM) that contains energy, reserve, and regulation markets. The VPP aggregates the wind farm (WF), photovoltaic power (PV), energy storage (ES), gas turbine (GT), and hydropower station (HS). Based on the uncertainty modeling for the output of uncontrollable power sources (UCPSs), such as the renewable energy in terms of WF and PV, this research develops countermeasures to reduce the penalty caused by the deviation between actual and predicted outputs of UCPSs. The other three controllable power sources (CPSs) are required to remain a certain reserve capacity for compensating the deviation to maximize the expected benefits of the whole VPP. By means of the quantile and superquantile theory, the proposed model considers the economic penalties beyond the reserve capacity and optimizes the allocation of reserve capacity to maximize the whole profit. With the construction of a mixed-integer nonlinear programming model, the best profits of the VPP in a variety of cases are reached and discussed. The experimental results demonstrate the effectiveness of diverse power sources integrated into a VPP, and the optimal bidding strategy of such renewable-based VPP in the DAM.

The main notation of this paper is stated above. Expressions “(t)” represent the value of the relevant variables at time t in the main text and are omitted in the nomenclature section for the sake of conciseness.

1. Introduction

To meet the ever-increasing needs for electricity and reduce the pollution of conventional thermal power plants, distributed energy resources (DERs) have been increased dramatically. Under the electricity market environment, it is worth exploring the issues related to the profits of DERs in the day-ahead market (DAM) that includes energy, capacity and ancillary service markets. The ancillary service markets can be further divided into the reserve market and regulation (up and down) market.

The virtual power plant (VPP), as a virtual aggregator of some interconnected power sources including the wind farms (WF), photovoltaic powers (PV), hydropower stations (HS), energy storages (ES), gas turbines (GT), etc., makes full use of these power sources to realize multi-energy complementation on the power supply side. VPP combines different types of power sources to be able to appear on the market as an independent power plant with a defined hourly output. By providing

energy and ancillary services to a system operator, VPP plays an important role in the electricity markets [1,2].

Regarding the bidding strategy of VPP in the DAM, Refs. [3–5] investigated a VPP consisting of some distributed generations (DGs), ES and demand response (DR) to participate in an energy market and a spinning reserve market. Benefitting from the thermal inertia of building and the complementarity of integrated energy, a VPP can provide spinning reserves by reducing the cooling supply of a building when the grid reserve capacity is scarce [6]. The optimization model constructed in the above-mentioned researches is a deterministic model, and thus the outputs of all power sources are assumed to be deterministic and completely available. These traditional methods may no longer be applicable to the VPPs that include the WF and PV with strong uncertainties.

The optimal bidding strategy considering the uncertainty of uncontrollable power sources (UCPSs) such as wind and photovoltaic power was investigated in previous work relevant to VPP. The related work [7] constructs a robust bidding model for VPP to participate in the day-ahead energy market, in which the operating costs of VPP is minimized while the renewable energy and market electricity price fluctuations are maximized and the availability of wind power is ensured by

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<https://doi.org/10.1016/j.ijepes.2022.108557>

Received 7 January 2022; Received in revised form 23 June 2022; Accepted 15 August 2022

Available online 7 September 2022

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Nomenclature		T_{bat}	Lifespan of the battery (year)
Acronyms		T_{HS}	Cost recovery cycle of the hydropower station (year)
CPS	Controllable power source	$\varphi^{up}/\varphi^{down}$	Up/ down penalty factor of positive/ negative power deviation between actual output and bidding of UCPS
DAM	Day ahead market	η	Charging or discharging efficiency of ES (p.u.)
E	Energy	κ	Multiple of the purchasing energy price divided by the selling energy price (p.u.)
ES	Energy storage	$\lambda^{su}/\lambda^{sd}$	Star-up/ shutdown cost of GT (\$)
GT	Gas turbine	Variables	
HS	Hydropower station	C_{chpur}	ES's charging cost through purchasing from the retail market (\$)
PV	Photovoltaic power	$E_{ESmin}/E_{ESmax}/E_{ES}(t)$	ES's minimum/maximum SOC and real time SOC at t o'clock (MWh)
RD	Regulation down	$f(x)$	The probability function of UCPS
RS	Reserve	$f^{-1}(\alpha)/f^{-1}(\bar{\alpha})$	Bilateral quantiles of given probability α and $\bar{\alpha}$ respectively
RU	Regulation up	M_2/M_3	Binary variable in the constraints
SOC	State of charge	n_s	Number of samples
UCPS	Uncontrollable power source	p^{up}/p^{down}	Up/down daily cost of the excess power beyond the reserve capacity (\$)
VPP	Virtual power plant	$p_{WF}/p_{PV}/p_{ES}/p_{GT}/p_{HS}$	Daily cost of the five categories of power sources (\$)
WF	Wind farm	p^{up}/p^{down}	Up/down excess power beyond the reserve capacity (MW)
Set and Indices		P_{chpur}	ES's charging power through purchasing from the retail market (MW)
m	Set of ancillary services markets	P_{chvpp}	ES's charging power from VPP (MW)
n	Set of up and down reserve capacity	P_{disch}	ES's discharging power (MW)
Φ	Set of UCPSs	$P_E/P_{RS}/P_{RU}/P_{RD}$	VPP's bidding capacity in energy, reserve, regulation up and regulation down markets (MWh)
t	Index of time in DAM	P_{GT}	Power of the gas turbine (MW)
Parameters		P_{HS}	Power of the hydropower (MW)
C_{bat}	Equivalent daily average battery cost of ES after discounting (\$/MW)	$P_{m,ES1}/P_{m,GT1}/P_{m,HS1}$	ES's/GT's/hydropower's bidding power in the ancillary service market in index m
C_{ESeom}	Operation and maintaining cost per unit energy of ES in life cycle (\$/MWh)	$P_{n,ES2}/P_{n,GT2}/P_{n,HS2}$	ES's/GT's/hydropower's reserve power for the UCPSs in index n
C_{ESpom}	Operation and maintaining cost per unit power of ES in life cycle (\$/MW)	P_{pre}/P_{act}	Prediction/actual value of UCPS (MW)
$C_{GT0}, C_{GT1}, C_{GT2}$	Constant, linear and quadratic terms of GT's cost function (\$, \$/MWh, \$/(MW ² h))	P_{PV}	Prediction value of photovoltaic power (MW)
$C_{Hsd,ins}$	Installed costs of hydropower (\$/MW)	P_{WF}	Prediction value of wind power (MW)
C_{HSom}	Operation and maintaining cost per unit power of hydropower (\$/MW)	res_{UCPS}	CPSs' reserve capacity for the deviation of UCPSs (MW)
C_{PV}	Photovoltaic power cost (\$/MWh)	u^{su}/u^{sd}	Binary variable = 1, if the GT is start-up/shutdown, and 0 otherwise
C_{WF}	Wind power cost (\$/MWh)	W_{ES}	Charging or discharging capacity of the ES (MWh)
E_{ESmax}/E_{ESmin}	Lower/ upper bound of SOC (MWh)	x	Prediction error of UCPS (MW)
p^E	Price in energy market (\$/MWh)	$\bar{x}$	The mean of samples (MW)
p^{RS}	Price in reserve market (\$/MWh)	Y^E/Y^m	Gross profit of the energy market and the ancillary service market in index m
p^{RU}/p^{RD}	Price in regulation up/down market (\$/MWh)	β	Confidence level
P_{ESmax}	Maximum charging/discharging power of ES (MW)	μ/σ	The mean and standard deviation of fitting normal distribution (MW)
P_{GTmax}/P_{GTmin}	Maximum/minimum power of GT in working condition (MW)		
$P_{GT, ramp}$	Ramping limits of GT (MW)		
P_{HSmax}	Maximum power of hydropower station (MW)		
P_{PVmax}	Maximum photovoltaic power (MW)		
P_{WFmax}	Maximum wind power (MW)		
Q_{HSmax}	Daily equivalent full generation hours of hydropower station (h)		
r	Discounting rate (p.u.)		
S_{bat}	Rated capacity of ES (MWh)		

the robust parameters. An optimal bidding model is constructed for the VPP to participate in the day-ahead and real-time markets by using a robust optimization method while considering the uncertainty of wind power in [8]. Robust optimization is also adopted in [9]. [10] proposes a stochastic adaptive robust optimization model. The uncertainty of wind power generation is represented by a confidence interval, and the uncertainty of electricity price is handled by a scenario model which is solved by a column and constraint generation algorithm (C&CG). At the same time, the literatures [11] and [12] also utilize the stochastic

optimization method to handle the optimal problem with the uncertainty of wind generation. The uncertainties of UCPSs have remarkable influences on the electricity market. Reference [11] stated that the generation companies need to pay the penalty cost for the deviation power of renewable energy in the real-time market. However, these researches did not explore the effects of controllable power sources (CPSs) in the VPP, which are beneficial to depressing the output fluctuations of UCPSs.

In fact, it is quite worthy to involve the interaction between CPSs and

UCPSs. On one side, the output deviations will cause UCPSs to be economically punished in the electricity market. Therefore, it is necessary to consider the economic impact brought by UCPSs and the countermeasures to reduce it. Meanwhile, CPSs in VPP such as the HS, GT, and ES have better responsive capacities, which can relieve the volatility and uncertainty of UCPSs' output, and reduce the penalty of VPP due to power deviations. Like in reference [13], the authors mentioned that the power deviations caused by volatile wind power productions can be mitigated by utilizing demand-side resources or battery storage. On the other side, the low cost of UCPSs energy supply can provide charging power for the ES. In other words, the two entities have inherent complementary advantages. In the studies associated with both WF and ES, a general method was presented to mitigate the fluctuation of the UCPS by configuring battery ES or pumped storage plant (PSP) to enhance its ability to participate in the DAM [14–16].

Reference [14] presented a normal distribution to capture the uncertainty of the forecasted wind power participating in the day-ahead energy market. The 95% confidence interval of the wind power forecast error distribution is drawn by selecting the quantiles to represent the fluctuating output. The under-generation and over-generation processes of wind power are stabilized by the discharging and charging operations of battery ES to achieve the purpose of increasing the revenue of the WF. Reference [15] ignored the uncertainty of wind power deviation but provided a predetermined deviation scenario, by which the power deviation of the WF can be reduced through the real-time correction effect of the PSP. By considering the forecast error of wind power as a probability distribution, a set of chance constraints are adopted in [16] to describe wind power scenarios, and the PSP is applied to suppress the deviation of WFs in the DAM. In this research, the ES provides the reserve capacity for wind farm only in the energy market, and it is difficult to handle the situation of VPP participating in multiple markets.

In summary, the previous studies related to the bidding strategy of VPP have not investigated how to mitigate the fluctuating outputs of UCPS with the aid of CPS in multiple markets. In these studies, the uncertainty arisen from UCPS are handled either by some parameters or by a confidence interval in a subjective way, and therefore the risks caused by exceeding the set parameters or beyond the confidence interval and the resulting economic penalties are not well quantified.

To overcome the deficiencies mentioned above, this paper presents the solutions for improvements. The reserve capacities of CPSs are objectively optimized under the constraints of economic penalty. The economic penalties of UCPSs and their economic optimization are integrated to find their best reserve capacities. A superquantile modeling concept, which is similar to that used in the risk assessment study [17,18], is introduced to characterize the expected value beyond the confidence interval in a target operation day. Based on the proposed models, the bidding strategy and the profitability of VPP are analyzed. The main contributions of this paper are threefold:

- (1) A novel mixed-integer nonlinear programming model is constructed for a VPP with WF, PV, ES, GT, and HS, to derive the effective bidding strategies of VPP in the energy, reserve, and regulation markets, i.e. multiple markets in the day ahead.
- (2) Aiming at depressing the power deviation of UCPS, an uncertainty modeling based on the probability distribution of prediction error is developed through utilizing the confidence intervals, which are determined by quantiles to characterize the reserve capacities provided by CPSs for UCPSs; in the model, the superquantile theory is adopted to determine the expected value beyond the quantiles and quantify its penalty cost.
- (3) Based on realistic data of power sources and markets, an in-depth analysis is conducted to illustrate the necessity of various power sources aggregated into a VPP and compare the profit rates of VPPs under different combinations, the findings of which can be

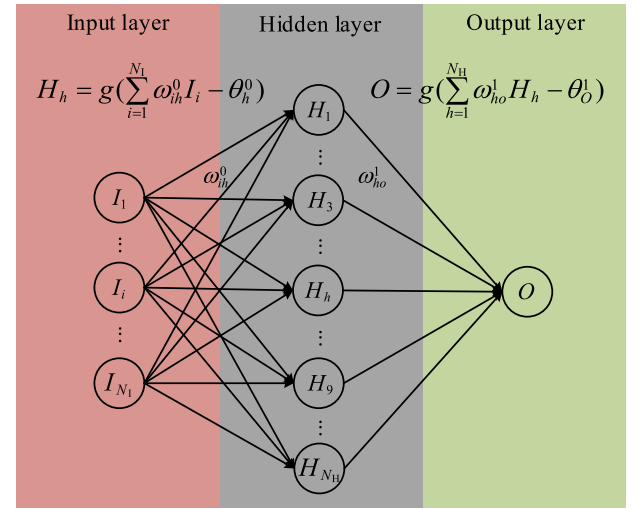


Fig. 1. The structure of the BP artificial neural network.

taken as the guidance for a diversity of power sources to form the VPP alliance.

The rest of this paper is organized as follows. Section 2 presents the UCPS's power prediction method in the day ahead, the selection of reserve capacity and the superquantile theory. Section 3 constructs the mixed-integer nonlinear programming model that consists of a nonlinear definite objective function and a set of linear convex constraints. Section 4 provides the results of case study based on the actual data. Section 5 offers several significant conclusions.

2. Uncertainty modeling of UCPSs

This section aims to introduce the forecast power of UCPSs participating in the DAM, and how to determine an interval from its forecast error distribution as the range of consideration for the reserve capacity. Subsequently, the introduction of superquantile theory characterizes the expected value of the random output of UCPSs beyond the reserve capacity.

2.1. Power prediction of UCPSs

In the research of this article, the generating capacity of UCPSs to participate in the electricity market is obtained by the point prediction of wind and photovoltaic power. Meanwhile, the CPSs' suppression effect to power imbalance of UCPSs relies on the power prediction error distribution information. Accurately predicting the power of UCPSs and obtaining a well-performed power prediction error distribution is, therefore, of great significance to the calculation of the expected revenue of the entire VPP. The generation forecast of a WF is taken as an example to illustrate the forecasting method and power prediction error distribution of UCPS. This approach is also applicable to handle the prediction of the PV, without repeating for brevity herein.

The UCPS possesses relatively large volatility and strong uncertainty. The utilization of prediction technology to access UCPSs power generation is more accurate than the traditional internal estimation according to the range of historical data. The participation of VPPs in the DAM requires certain generation values of 24 h. From the time scope, it is a short-term forecast. The artificial neural network (ANN) algorithm is one of the widely used learning algorithms, inspired by biological neural networks, which is capable of nonlinear approximation-oriented forecasting. Based on the different connection weights and thresholds among neurons in layers, it can approximate any nonlinear system in theory. Therefore, when dealing with such highly uncertain nonlinear

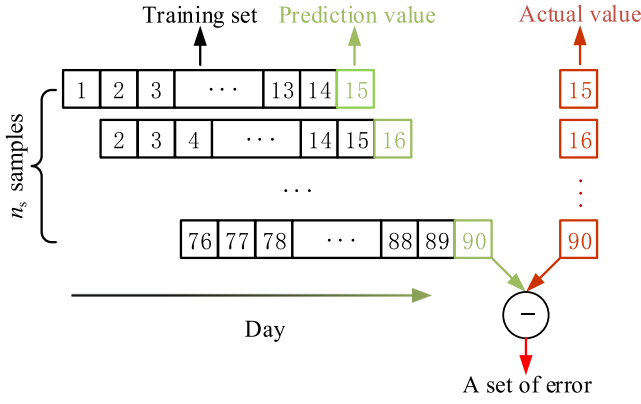


Fig. 2. The schematic diagram of rolling prediction to generate errors.

data for the wind power, adoption of the ANN algorithm for forecasting brings advantages, which promote the broad utilization within short-term prediction [19].

The BP artificial neural network is selected to predict the power of a WF in Texas. In this study, the prediction model is trained by a three-layer ANN with one input, hidden and output layer, respectively [14]. The structure of the network can be depicted in Fig. 1.

In this figure, I_i , H_h and O represent the data in input, hidden and output nodes, respectively. I_{N_i} and H_{N_h} are the node quantities of input and hidden layers. ω_{ih}^0 and ω_{ho}^1 are the weights between the connected nodes. While the θ_h^0 and θ_o^1 are the threshold value of hidden layer and output layer. The equations between different layers are the activation functioning(y), which can be calculated as follows:

$$g(y) = \frac{1}{1 + e^{-y}} \quad (1)$$

The input layer includes 5 neuron nodes, corresponding to the prediction meteorological parameters affecting wind power generation, which include the wind speed, wind direction, temperature, humidity, and atmospheric pressure in each hour. The numerical weather prediction information can be obtained from the meteorological bureau. The output layer has 1 neuron node, which is the wind power generation, and the number of hidden layer neurons is set to 11. The trainlm and purelin functions are selected as the training function and node transferring function respectively in the model training process.

2.2. Uncertainty from prediction error

Due to the existence of the prediction error, the market will impose the penalty on the VPP considering such deviation between the actual power value and the bidding value if the prediction result is directly used as the bidding capacity of the WF in the DAM without any countermeasure. For better grasping the prediction error, the normal distribution and its formation process are introduced in Fig. 2.

This figure presents the rolling process of generating prediction errors based on the data in a quarter (i.e. 90 days). The generation data on a certain day is used as the testing set (the green square), meanwhile, these meteorological data of every hour in the past 14 days before that are the training set (the black square). Every prediction value is forecasted by the model trained by the past 14 days. Particularly, for each prediction day, the ANN model is required to update based on the latest 14 days. The last day of the first quarter, March 31, is selected as the operation day in the market, thus the bidding strategy needs to be scheduled on March 30.

By analyzing the error between the prediction generation value and the actual generation value (the red square) of the historical data for a long period of time (76 days), the probability distribution satisfied by the set of error on the prediction day can be described. Considering the

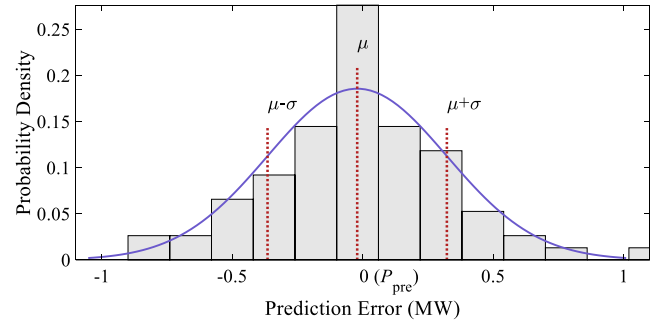


Fig. 3. Prediction error frequency histogram of wind power at 5 a.m.

forecast error for each hour of the following day is different, while the electricity price at different moments changes as well, the error of each hour should be separately counted in order to accurately calculate the expected revenue. The process of estimating normal distribution parameters is given as follows:

$$X = P_{\text{pre}} - P_{\text{act}} \quad (2)$$

where X defines the prediction error vector. P_{pre} and P_{act} denote the prediction and actual generation value vectors of the UCPS. After obtaining a set of prediction errors through (2), the unbiased estimator of the mean and variance of the normal distribution can be calculated by the following formulas:

$$\mu = \frac{1}{n_s} \sum_{i=1}^{n_s} x_i \quad (3)$$

$$\sigma^2 = \frac{1}{n_s - 1} \sum_{i=1}^{n_s} (x_i - \bar{x})^2 \quad (4)$$

where x_i ($i = 1, 2, \dots, n_s$) defines the i th element in vector X . Variables μ and σ denote the mean and standard deviation of the fitting normal distribution, respectively. n_s is the number of the samples and $\bar{x}$ is the mean value of samples (i.e. μ). Through the parameter estimation of the normal distribution. The probability function of wind power prediction error can be expressed as below:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (5)$$

where x denotes the variable of prediction error and $f(x)$ is the probability function. In Fig. 3, the data at 5 a.m. is taken as an example to depict the frequency histogram of prediction error for wind power.

Fig. 3 indicates that the calculated normal distribution more or less exactly fits the prediction error probability of wind power. In the figure, when the prediction error is 0, it corresponds to the point prediction of wind power P_{pre} on the bidding day.

2.3. Reserve capacity for UCPS

Based on the distribution, combined with the quantiles of the normal distribution, the fluctuation interval of wind power at a certain confidence level can be drawn:

$$L(\beta) = [P_{\text{pre}} + f^{-1}(\alpha), P_{\text{pre}} + f^{-1}(\bar{\alpha})] \quad (6)$$

$$\bar{\alpha} - \alpha = \beta \quad (7)$$

where the $L(\beta)$ portrays the dynamic internal based on the prediction value P_{pre} adding with quantiles. $P_{\text{pre}} + f^{-1}(\alpha)$ and $P_{\text{pre}} + f^{-1}(\bar{\alpha})$ are the lower and upper limits of the internal by using inverse functions $\text{off}(x)$. And equation (7) connects the given probability value, α and $\bar{\alpha}$, with the confidence level β . For example in Fig. 3, an internal $[\mu - \sigma, \mu + \sigma]$ indicates the most likely wind power fluctuation range under confidence level 0.6826. The upper and lower bound of such interval corresponds to

the reserve capacity (down and up) provided by CPSs for UCPSs. Meanwhile, $\underline{\alpha}$ and $\bar{\alpha}$ are equal to 0.1587 and 0.8413 respectively. The VPP will not be punished if actual outputs of UCPSs locate in the internal $[\mu - \sigma, \mu + \sigma]$ under the compensation of reserve capacity. Instead, the VPP will pay the penalty for the deviation capacity if actual outputs of UCPSs fall beyond the scope of $[\mu - \sigma, \mu + \sigma]$.

Considering the quantiles are selected as the reserve capacity by CPSs for UCPSs, the superquantiles are the first time introduced to depict the expected power value beyond the reserve capacity (i.e. the quantiles). The superquantiles calculation of the wind and solar power can be expressed as follows [17]:

$$P^{\text{sup}}(t) = \sum_{\Phi} \frac{\int_{x \geq f_{\Phi(t)}^{-1}(\bar{\alpha})} x f_{\Phi(t)}(x) dx}{1 - \bar{\alpha}} \quad (8)$$

$$P^{\text{sqdown}}(t) = \sum_{\Phi} \frac{\int_{x \leq f_{\Phi(t)}^{-1}(\underline{\alpha})} x f_{\Phi(t)}(x) dx}{\underline{\alpha}} \quad (9)$$

where the set $\Phi = \{\text{WF}, \text{PV}\}$ includes the UCPSs in terms of WF and PV. $P^{\text{sup}}(t)$ and $P^{\text{sqdown}}(t)$ are the up superquantile and down superquantile in the time period t . In addition, the symbol “ (t) ” represent the variables in time period t and will not be explained repeatedly in the rest of this paper for the sake of conciseness. Based on (8) and (9), the difference value of superquantiles and quantiles are employed to calculate the penalty capacity in (10) and (11):

$$P^{\text{up}}(t) = P^{\text{sup}}(t) - \sum_{\Phi} f_{\Phi(t)}^{-1}(\bar{\alpha}) \quad (10)$$

$$P^{\text{down}}(t) = \sum_{\Phi} f_{\Phi(t)}^{-1}(\underline{\alpha}) - P^{\text{sqdown}}(t) \quad (11)$$

where $P^{\text{up}}(t)$ and $P^{\text{down}}(t)$ are the penalty capacity, which mean the excess capacity and insufficient capacity compared with the scope of reserve capacity respectively and will be punished financially in the electricity market. From the structure of the formulas (10) and (11), it is not difficult to notice the definite integral operation, which makes the whole programming model intractable. Due to that, an appropriate discrete set of values $\{0, 1\sigma, 2\sigma, 3\sigma\}$ are possibly selected as the reserve capacity by CPSs for UCPSs, res_{UCPS} , and results of sensitivity analysis will be given later to pursue the best res_{UCPS} . According to the 3σ criterion in the statistical theory, the values of a few of such discrete reserve capacities respectively correspond to $\underline{\alpha} = \bar{\alpha} = \{0.5, 0.1587, 0.0228, 0.0013\}$. Once determining the values of $P^{\text{up}}(t)$ and $P^{\text{down}}(t)$, independently, the solvability of the model is guaranteed. Through the theory above, the prediction power of the UCPSs, the reserve capacity supplied by CPSs for the UCPSs, and the penalty capacity beyond the reserve capacity can be obtained.

2.4. Summary of uncertainty handling

Section 2 firstly forms a prediction error distribution based on multiple sets of prediction errors, which can represent the probabilistic characteristics of prediction errors. The reserve capacity selected in this paper corresponds to the quantile of a certain confidence interval. When the forecast errors of wind power and photovoltaic power generation are within this confidence interval, there will be no penalty cost. For the prediction errors outside the confidence interval, this study uses two-sided superquantiles to determine the expected value of the uncertain error, which in turn generates a corresponding penalty cost. In addition, the penalty cost will be an important factor discussed in the next section.

3. Model of optimal bidding strategy

This section introduces the model of the optimal bidding strategy for a VPP, which participates in the energy, reserve, regulation up and regulation down markets. The connections between the optimal bidding

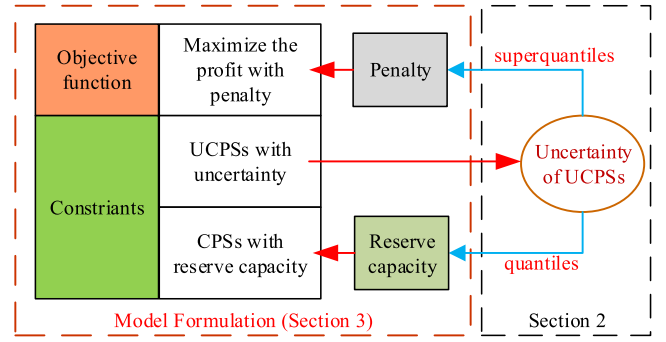


Fig. 4. The structure of the model with considering reserve capacity.

strategy model formulated in Section 3 and the theories of uncertainty in Section 2 can be drawn as follows:

Fig. 4 shows that the penalty and the reserve capacity model constructed in Section 2 will be further drawn into Section 3 to formulate the bidding strategy model. In the rest of this section, the model formulation can be divided into several parts. First of all, the objective function, i.e. the net profit, of the VPP is formulated, and then the constraints that various power sources are subject to will be introduced correspondingly.

3.1. Objective function of the model

Accurate analysis of the expected profits of VPP relies on the refined cost models of power sources in VPP. Therefore, this part firstly discusses the daily costs of the WF, PV, ES, GT, and HS in the DAM, and the penalty costs of capacity imbalance. In addition, since the time resolution is in hours, in terms of value, the scheduled generation power data (MW) is equal to the bidding capacity (MWh). It is not indispensable to strictly distinct the units MW and MWh in the calculation process.

(1) Daily cost of the UCPSs

The wind power and photovoltaic power generation have the characteristics of zero marginal cost, so one-time investment cost combined with depreciation rate to calculate their levelized cost in the life cycle is generally considered to form the cost expression in the current study. Therefore, the daily cost model of the power generation from WF and PV in the DAM can be simply expressed as the following formula:

$$P_{\text{WF}} = \sum_{t=1}^{24} P_{\text{WF}}(t) C_{\text{WF}} \quad (12)$$

$$P_{\text{PV}} = \sum_{t=1}^{24} P_{\text{PV}}(t) C_{\text{PV}} \quad (13)$$

where the C_{WF} and C_{PV} denote the generation cost of UCPSs per MWh, which are available in [20] and [21]. $P_{\text{PV}}(t)$ and $P_{\text{WF}}(t)$ are the generating power of the PV and WF in the time period. The daily costs of the two types of renewable energy can be expressed by p_{WF} and p_{PV} , respectively.

(2) Daily cost of the ES

Due to the characteristics of battery consumption, the cost of the ES system mainly includes the cost of one-time purchasing of batteries, the operation & maintenance cost of the ES power station, and the cost of charging from the grid. Therefore, the cost function can be formulated as follows:

$$P_{\text{ES}} = C_{\text{bat}} + \sum_{t=1}^{24} [C_{\text{ESom}}(t) + C_{\text{chpur}}(t)] \quad (14)$$

where the C_{bat} is the equivalent fixed cost for bidding day and the one-time installed cost can be obtained from the data in the latest project bidding, $C_{\text{ESom}}(t)$ denotes the operation and management cost of ES for each hour, and $C_{\text{chpur}}(t)$ is the cost of charging from the market. C_{bat} can be calculated from present worth to equivalent annual cost as follows:

$$C_{\text{bat}} = \frac{C_{\text{bat,ins}}}{365} \cdot \frac{r(1+r)^{T_{\text{bat}}}}{(1+r)^{T_{\text{bat}}} - 1} \quad (15)$$

where $C_{\text{bat,ins}}$ denotes the installed battery cost of the ES and r is the discount value. In addition, T_{bat} is the lifespan of the ES. A popular model for calculating $C_{\text{ESom}}(t)$ is presented in [22] as follows:

$$C_{\text{ESom}}(t) = \frac{1}{8760T_{\text{bat}}} \left[C_{\text{ESpom}} P_{\text{ESmax}} \frac{(1+r)^{T_{\text{bat}}} - 1}{r(1+r)^{T_{\text{bat}}}} + C_{\text{EScom}} W_{\text{ES}}(t) \sum_{T=1}^{T_{\text{bat}}} (1+r)^{-T} \right] \quad (16)$$

where C_{ESpom} and C_{EScom} are the operation & maintenance cost per unit power and per unit energy of the ES, respectively. $W_{\text{ES}}(t)$ represents the charging or discharging capacity of the ES in the time period t . The P_{ESmax} is the maximum charging or discharging power of the ES. Another variable $C_{\text{chpur}}(t)$ can be calculated as below:

$$C_{\text{chpur}}(t) = P_{\text{chpur}}(t) \kappa_q p^E(t) \quad (17)$$

where the κ_q is the ratio of the retail market price to the energy market price in the q^{th} quarter. According to the information from EIA (Energy Information Administration), the average ratio in the first quarter of 2019 was 1.45 [23–24]. $p^E(t)$ is the energy market price while $P_{\text{chpur}}(t)$ denotes the charging power through purchasing from the retail market.

(3) Daily cost of the GT

The daily cost function of the GT generation is similar to conventional thermal power plants, which can be expressed by a quadratic function. In addition, start-up and shutdown costs should be covered:

$$P_{\text{GT}} = \sum_{t=1}^{24} \{ \text{sign}[P_{\text{GT}}(t)] \cdot C_{\text{GT0}} + C_{\text{GT1}} P_{\text{GT}}(t) + C_{\text{GT2}} P_{\text{GT}}^2(t) + \lambda^{\text{su}} u^{\text{su}}(t) + \lambda^{\text{sd}} u^{\text{sd}}(t) \} \quad (18)$$

where function $\text{sign}(\cdot)$ indicates the status of GT (equals to 1 when the GT is operating and 0 otherwise). $P_{\text{GT}}(t)$ denotes the generating power of GT. C_{GT0} , C_{GT1} and C_{GT2} are the constant, linear, and quadratic terms of the GT's cost function (\$/MWh, \$/(MW²·h)), respectively. $u^{\text{su}}(t)$ and $u^{\text{sd}}(t)$ are two binary variables to manifest the start-up and shut-down status. λ^{su} and λ^{sd} are the cost of the GT each time. The fuel price per unit can be identified from the EIA [25], and fuel consumption parameters can be accessible in [26].

(4) Daily cost of the HS

On the grounds that the hydro turbine has a lower expense when starting up or shutting down, the constructive cost model of the HS is formulated with the operation, maintenance, and equivalent fixed cost for bidding day:

$$p_{\text{HS}} = \sum_{t=1}^{24} [C_{\text{HSom}} P_{\text{HS}}(t)] + \frac{P_{\text{HSmax}} C_{\text{HSd,ins}}}{365} \cdot \frac{r(1+r)^{T_{\text{HS}}}}{(1+r)^{T_{\text{HS}}} - 1} \quad (19)$$

where the C_{HSom} and $C_{\text{HSd,ins}}$ are the operation & maintenance and the installed cost of the HS. $P_{\text{HS}}(t)$ is the generating power of the HS. Due to the long-life cycle, the lifespan of the HS varies in different projects and it is difficult to gauge. The T_{HS} is selected as the cost recovery cycle for 20 years, and the installed cost can be found in [27].

(5) Daily penalty cost of the capacity deviation

If the output of the UCPS deviates far away from the scheduled value, it will affect the secure operation of the power system. Due to the existence of UCPSs in the VPP, the daily costs should consider not only operation, maintenance, and the fixed costs of various power sources, but also the economic penalties caused by excess or shortage of UCPSs in the market penalty, which is the first time to consider this sort of situation. Combining with equations (10) and (11), the daily penalty cost model can be presented by equations (20) and (21):

$$p^{\text{up}} = \sum_{t=1}^{24} \varphi^+ p^E(t) P^{\text{up}}(t) \quad (20)$$

$$p^{\text{down}} = \sum_{t=1}^{24} \varphi^- p^E(t) P^{\text{down}}(t) \quad (21)$$

where $p^E(t)$ is the electricity price in energy market. The variables p^{up} and p^{down} are the penalties correspond to the excessive and shortage generating capacity. The cost of punishment can be implemented based on the settlement price of the energy market and the penalty factors φ^+ and φ^- , which are set to 1 and 2, respectively.

(6) Total daily cost of the VPP

Based on the construction of the above model, the cost model of the entire VPP in the DAM is combined as follows:

$$p_{\text{vpp}} = p_{\text{WF}} + p_{\text{PV}} + p_{\text{ES}} + p_{\text{GT}} + p_{\text{HS}} + p^{\text{up}} + p^{\text{down}} \quad (22)$$

where the p_{vpp} is the daily cost of the whole VPP. The objective function of the optimal bidding model for the VPP to participate in the day-ahead market is aiming to maximize the benefits:

$$\max Y^E + Y^{\text{RS}} + Y^{\text{RU}} + Y^{\text{RD}} - p_{\text{vpp}} \quad (23)$$

The variables Y with superscript E, RS, RU, and RD are the gross profit in the energy, reserve, regulation up, and regulation down markets. The detailed income determination of each market can be cast below:

$$Y^{\mathfrak{N}} = \sum_{t=1}^{24} p^{\mathfrak{N}}(t) P_{\mathfrak{N}}(t), \mathfrak{N} \in \{E, \text{RS}, \text{RU}, \text{RD}\} \quad (24)$$

where the $p^{\mathfrak{N}}(t)$ and $P_{\mathfrak{N}}(t)$, $\mathfrak{N} \in \{E, \text{RS}, \text{RU}, \text{RD}\}$ denote the prices and bidding capacities of different markets. At this point, the construction for the objective function considering VPP's participation in the DAM is accomplished.

3.2. Constraints of the model

Due to the unchangeable prediction power of UCPSs, they are just granted to engage in energy market. While the CPSs are allowed to participate both in the energy and ancillary markets. The constraints of the bidding strategy model are divided into several parts: 1) the constraints of the VPP in general, 2) the constraints of the ES, 3) the constraints of the GT, 4) the constraints of the HS, and 5) the constraints of the reserve capacity from CPSs for UCPSs.

(1) Constraints of the VPP in general

The constraints mainly include the relationship between the bidding capacity of VPP to participate in various markets and the scheduled capacity of each power source inside.

$$0 \leq P_{\text{E}}(t) = P_{\text{WF}}(t) + P_{\text{PV}}(t) + P_{\text{disch}}(t) + P_{\text{GT}}(t) + P_{\text{HS}}(t) - P_{\text{chvpp}}(t) \quad (25)$$

$$0 \leq P_{\text{RS}}(t) = P_{\text{RS,ES}}(t) + P_{\text{RS,GT}}(t) + P_{\text{RS,HS}}(t) \quad (26)$$

$$0 \leq P_n(t) = P_{n,ES1}(t) + P_{n,GT1}(t) + P_{n,HS1}(t), \quad n \in \{RU, RD\} \quad (27)$$

$$0 \leq P_{m,ES}(t), P_{m,GT1}(t), P_{m,HS1}(t), m \in \{RS, RU, RD\} \quad (28)$$

where the $P_E(t)$ and $P_m(t)$, $m \in \{RS, RU, RD\}$ are the energy and ancillary markets bidding capacity of the overall VPP. $P_{disch}(t)$ and $P_{chvpp}(t)$ are the discharging capacity and charging capacity from the VPP inside of ES. The variables $P_{RS,ES}(t)$, $P_{RS,GT}(t)$ and $P_{RS,HS}(t)$ represent the scheduled reserve market capacities of CPSSs, separately; whilst $P_{n,ES1}(t)$, $P_{n,GT1}(t)$ and $P_{n,HS1}(t)$ are the scheduled regulation markets capacities of CPSSs, individually. Equation (25) indicates that the capacity of a VPP to participate in energy market equals to the sum of the generation of various power sources. Equations (26) and (27) manifest the ancillary markets VPP participated in include the scheduled capacities of the ES, GT, and HS. In addition, all the variables are subject to non-negative restrictions in (28).

(2) Constraints of the ES

Power and capacity constraints of the ES are integrally included:

$$P_{ch}(t) = P_{chvpp}(t) + P_{chpur}(t) \quad (29)$$

$$0 \leq P_{ch}(t) \leq [1 - M_1(t)]P_{ESmax} \quad (30)$$

$$0 \leq P_{disch}(t) \leq M_1(t)P_{ESmax} \quad (31)$$

$$M_1(t) = 0, 1 \quad (32)$$

$$W_{ES}(t) = P_{ch}(t) + P_{disch}(t) \quad (33)$$

$$0 \leq P_{n,ES2}(t), n \in \{RU, RD\} \quad (34)$$

$$0 \leq P_{disch}(t) + P_{RS,ES}(t) + P_{RU,ES1}(t) + P_{RU,ES2}(t) \leq P_{ESmax} \quad (35)$$

$$0 \leq P_{ch}(t) + P_{RD,ES1}(t) + P_{RD,ES2}(t) \leq P_{ESmax} \quad (36)$$

$$-P_{ESmax} \leq P_{disch}(t) - P_{RD,ES1}(t) - P_{RD,ES2}(t) \quad (37)$$

$$E_{ES}(t) + \eta[P_{chpur}(t) + P_{chvpp}(t)] - \frac{1}{\eta}P_{disch}(t) = E_{ES}(t+1) \quad (38)$$

$$0 \leq \frac{1}{\eta}[P_{disch}(t) + P_{RS,ES}(t) + 0.25P_{RU,ES1}(t) + P_{RU,ES2}(t)] \leq E_{ES}(t) - E_{ESmin} \quad (39)$$

$$0 \leq P_{RS,ES}(t) + 0.25P_{RU,ES1}(t) + P_{RU,ES2}(t) \leq \zeta E_{ESmax} \quad (40)$$

$$0 \leq 0.25P_{RD,ES1}(t) + P_{RD,ES2}(t) \leq \zeta E_{ESmax} \quad (41)$$

$$E_{ESmin} \leq \left[E_{ES}(t) - \frac{1}{\eta}P_{disch}(t) + 0.25\eta P_{RD,ES1}(t) + \eta P_{RD,ES2}(t) \right] \leq E_{ESmax} \quad (42)$$

$$E_{ESmin} \leq E_{ES}(t) \leq E_{ESmax} \quad (43)$$

$$E_{ES}(24) = E_{ES}(0) \quad (44)$$

where $P_{ch}(t)$ is the charging capacity of ES. The $M_1(t)$ is the binary variable to restrict the charging or discharging status of ES. The variables $P_{n,ES2}(t)$, $n \in \{RU, RD\}$ are the reserve up and reserve down capacity provided by ES for UCPSSs. E_{ESmin} and E_{ESmax} are the minimum and maximum SOC of ES. $E_{ES}(t)$ is the real-time SOC at t o'clock. η denotes the charging and discharging efficiency of ES, and ζ indicates the quantity share of the ES for the reserve to charging or discharging, as a fraction of maximum SOC, which can be 0.1 according to [28]. The constraints of ES's charging or discharging operation are expressed in (29)-(32), while the simultaneous charging and discharging are prohibited. Equation (34) ensures the non-negativity of the reserve capacity

by ES for UCPSSs. Equation (35) indicates that the sum power in three kinds of market of ES, and the reserve up capacity for the UCPSSs cannot exceed the maximum rated power. Similarly, the formula (36) describes that the sum power in two kinds of market of the ES, and the reserve down capacity for the UCPSSs cannot exceed the maximum rated power. On the basis of discharging in the energy market, the power of the ES participating in the regulation down market, and the charging capacity reserved for UCPSSs cannot surpass the maximum charging power of the ES is illustrated in (37). Constraints (38)-(43) combine the SOC to further limit the capacity of participating in each type of market. The coefficient 0.25 is the equivalent conversion of the regulation market scheduling time of 15 min. The equation (44) restricts the beginning and ending SOC in the day.

(3) Constraints of the GT

$$M_2(t)P_{GTmin} \leq [P_{GT}(t) + P_{RS,GT}(t) + P_{RU,GT1}(t) + P_{RU,GT2}(t)] \leq M_2(t)P_{GTmax} \quad (45)$$

$$M_3(t)P_{GTmin} \leq [P_{GT}(t) - P_{RD,GT1}(t) - P_{RD,GT2}(t)] \leq M_3(t)P_{GTmax} \quad (46)$$

$$M_2(t), M_3(t) = 0, 1 \quad (47)$$

$$0 \leq P_{n,GT2}(t), n \in \{RU, RD\} \quad (48)$$

$$-P_{GT,ramp} \leq P_{GT}(t+1) - P_{GT}(t) \leq P_{GT,ramp} \quad (49)$$

where the $M_2(t)$ and $M_3(t)$ are independent binary variables. Since the GT has the minimum technical output, equation (45) indicates that the sum of generating capacity, reserve capacity, regulation up capacity, and the reserve up capacity for UCPSSs is limited between the maximum and minimum technical output, or equals to 0. The variables P_{GTmin} and P_{GTmax} are the minimum and maximum generation power of the GT under the operation status. $P_{n,GT2}(t)$, $n \in \{RU, RD\}$ are the reserve up and reserve down capacity provided by the GT for UCPSSs. The variable $P_{GT,ramp}$ is the ramping limits of the GT. Similarly, (46) also restricts the sum of generating capacity, regulation down capacity, the reserve down capacity for UCPSSs within the limits. Equation (48) denotes the non-negative constraint containing the reserve up and reserve down capacity reserved by the GT for UCPSSs, and equation (49) is the ramping constraint of GT.

(4) Constraints of the HS

$$0 \leq P_{HS}(t) + P_{RS,HS}(t) + P_{RU,HS1}(t) + P_{RU,HS2}(t) \leq P_{HSmax} \quad (50)$$

$$0 \leq P_{HS}(t) - P_{RD,HS1}(t) - P_{RD,HS2}(t) \quad (51)$$

$$0 \leq P_{n,HS2}(t), n \in \{RU, RD\} \quad (52)$$

$$\sum_{t=1}^{24} P_{HS}(t) = Q_{HSmax1} \quad (53)$$

where Q_{HSmax1} is the daily equivalent full generation hours of the HS affected by water sources condition in the first quarter, which is available from [29] and [30]. $P_{n,HS2}(t)$, $n \in \{RU, RD\}$ are the reserve up and reserve down capacity provided by the hydropower for UCPSSs. Equation (50) indicates that the HS's participation in energy, reserve, regulation up, and reserve up capacity for UCPSS is reasonably less than the maximum output of the HS. In the same way, the bidding capacities in energy, regulation down markets, and the reserve up capacity for UCPSSs are within limits in (51). Equation (52) is a non-negative constraint on the reserve up and reserve down capacities of the hydropower reserved for UCPSSs. Though the power generation capacity of the HS is affected by water sources condition in a day, it can be flexibly adjusted within a certain time period. Equation (53) expresses the restriction on the total hydropower generation capacity in a day.

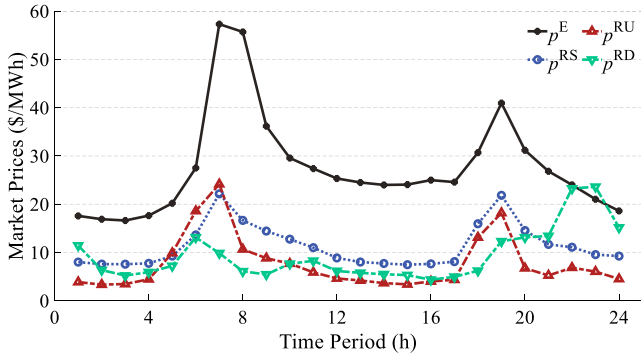


Fig. 5. Average prices of various electricity markets.

(5) Constraints of the reserve capacity for UCPSs

It is worth noting that when the actual power of the UCPSs is lower than the bidding capacity, the CPSs are required to increase their generation capacity. Conversely, when the actual power of the UCPS is higher than the bidding capacity, the CPSs are required to decrease their generation capacity. The interactions relating to the reserve capacity can be formulated with the following equality constraints as:

$$P_{RU,ES2}(t) + P_{RU,GT2}(t) + P_{RU,HS2}(t) = \sum_{\Phi} -f_{\Phi(i)}^{-1}(\underline{\alpha}) \quad (54)$$

$$P_{RD,ES2}(t) + P_{RD,GT2}(t) + P_{RD,HS2}(t) = \sum_{\Phi} f_{\Phi(i)}^{-1}(\bar{\alpha}) \quad (55)$$

Based on the work above, the optimal bidding strategy model for the VPP to participate in the DAM including the energy, reserve, and regulation markets has been finally constructed. This model incorporates these from equations (12) to (55).

4. Case study

To evaluate the performance of the proposed model, some critical case studies based on the practical data are analyzed. In addition, all the optimal models in the main and comparative experiments are solved in Matlab using Yalmip with Cplex as the solver.

In the ERCOT[24], electricity market in Texas, the VPP operator firstly form the distribution based on multiple sets of prediction errors. Based on the prediction error distribution parameters, the bidding plan can be solved through the constructed optimal model. Then the VPP operator submit the bidding plan in DAM before 16:00 based on the optimal results. And the energy, reserve, regulation down and up markets were cleared at 18:00, then the winning quantity in each market of the VPP can be settled. In the real-time operational process, the operator will dispatch the ancillary service if needed. The VPP will get paid through the winning power in ancillary market whether or not the power is used because of the opportunity cost.

But the servicing duration of reserve and regulation market is different. Specifically, the reserve service provides one-hour duration each time, while the regulation service provides 15-minutes duration each time. Hence, when the power in the two markets is same, the reward of the latter is 1/4 of the former due to the difference in MWh.

4.1. Basic data

In order to scrutinize the benefit comparison of various power sources, it's well-reasoned to unify the installed capacity of power sources into a same quantitative value. By uniformly scaling the generation data of the WF, the 16 MW installed capacity of the WF is logically converted to a well-compared value, 10 MW. The same approach is adopted to scale the installed capacity of the PV. The

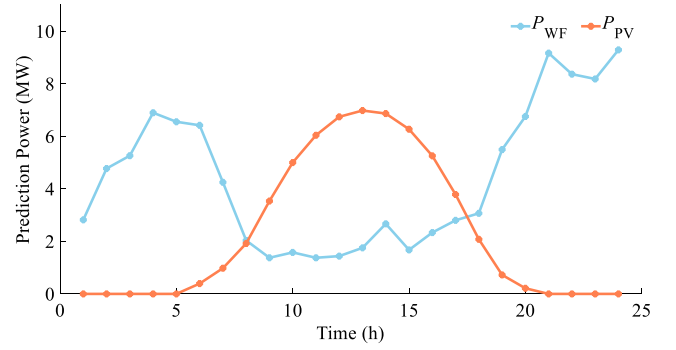


Fig. 6. Forecasted wind power and photovoltaic power.

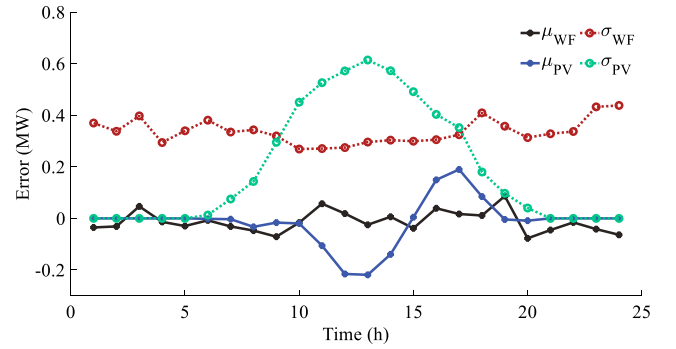


Fig. 7. Normal distribution parameters of UCPSs' prediction error.

installed capacities of all CPSs are set to 10 MW, and the maximum SOC of the ES is 60 MWh. The other detailed parameters of all these power sources are clearly attached in Table A1 in the appendix.

Due to the daily similarity of electricity prices, electricity price data under different conditions will only affect the absolute benefit value (viz: the optimal value of the model) and have little impact on the bidding strategy (viz: the optimal solution). Therefore, the electricity price data adopts the first-quarter average value in Texas's markets in the year of 2019 (see Fig. 5), which can be collected from [24]:

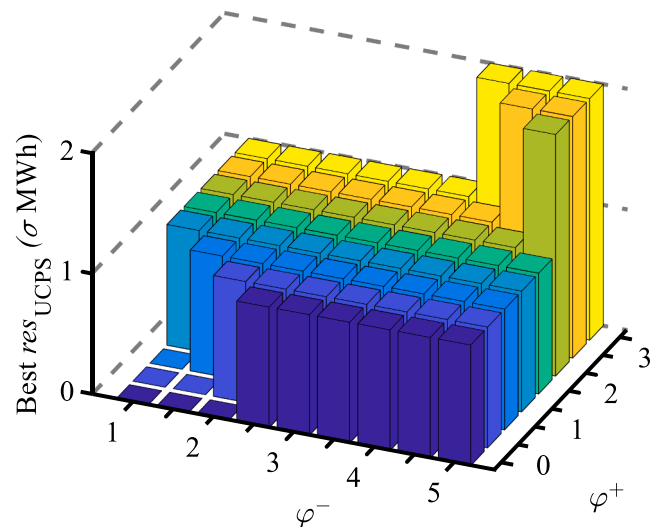


Fig. 8. Sensitivity analysis of optimal reserve capacity under different penalty coefficients.

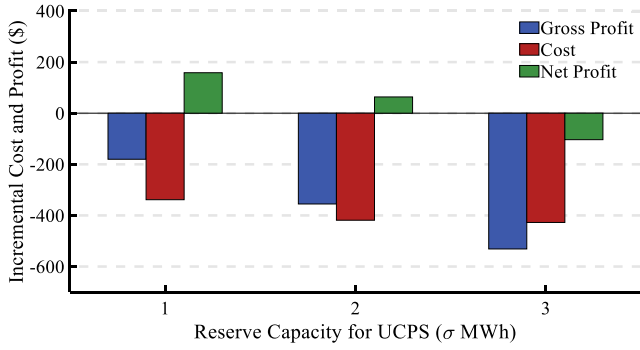


Fig. 9. Economic comparison of different reserve capacities compared to the one without reserve capacity.

4.2. Power prediction of UCPSs

The 90-day data of a WF and a PV in Texas from January 1 to March 31 in the year of 2019 are studied. The method in Section 2 is used to obtain the predicted power of UCPSs on the bidding day (April 1st), as well as the normal distribution parameters of the prediction error, which are illustrated in Fig. 6 and Fig. 7, individually:

4.3. Sensitivity analysis of CPSs' reserve capacity for UCPSs

Considering the calculation of the superquantiles in the constructed optimization model involves the integral form, it cannot be tackled directly by the optimal solver. Therefore, a method mentioned in Section 2.2 is put forward to handle it. First, a deterministic reserve capacity is tentatively selected and its corresponding superquantiles can be calculated offline. Then the deterministic and tractable optimization model is solved with one shot. Last but not least, by repeatedly solving the optimization model under different reserve capacities, the best reserve capacity can be found finally.

On account of the best reserve capacity is directly related to the economic penalty received by the VPP, the numerical sensitivity analysis method is employed herein to investigate the relationship between the optimal reserve capacity and the output deviation penalty coefficient, which has been shown in Fig. 8. The changing step size of the penalty coefficient is set to 0.5 with consideration of the computation burden. The penalty coefficient φ^+ changes from -0.5 to 3 , whilst φ^- varies from 1 to 5 .

It can be found that when the penalty coefficients are relatively small ($\varphi^+ + \varphi^- \leq 1.5$), CPSs will not prepare reserve capacity for UCPSs; but when the penalty coefficients are extremely large ($\varphi^+ + \varphi^- \geq 7$), CPSs tends to provide 2σ MWh reserve capacity. In most penalty coefficient cases ($2 \leq \varphi^+ + \varphi^- \leq 6.5$), the best reserve capacity provided by CPSs for UCPSs is σ MWh.

An instance in Fig. 9, based on the $\varphi^+ = 1$ and $\varphi^- = 2$ is shown to illustrate the profit difference based on various reserve capacities compared to the situation without reserve capacity. When VPP adopts to reserve capacity for UCPSs, its gross profit decreases due to the tighter power and capacity constraints of CPSs. But the cost reduces as a consequence of cheaper penalty with the help of the reserve capacity. Through experiments, it is found that when the reserve capacity is $0, \sigma, 2\sigma$, and 3σ , the penalty costs of VPP are $438.44 \$$, $91.59 \$$, $9.3517 \$$, and $0.40445 \$$, respectively. Thus, the changes in net profit (gross profit minus cost) appear two different trends. Too much reserve capacity, e.g. 3σ MWh, limits its own bidding strategy, namely, the feasible region in the optimization problem, resulting in the expected net profit not being as good as the situation without reserve capacity. When the reserve capacity is 1σ MWh or 2σ MWh, the total expected net profit will be higher than the one without the involvement of reserve capacity. Particularly, the net profit with a reserve capacity of 1σ MWh is

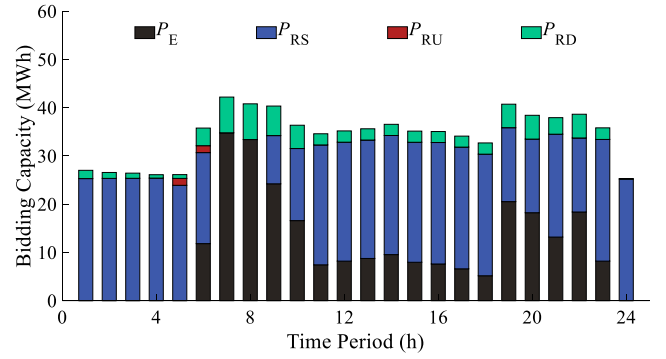


Fig. 10. VPP's bidding strategy in energy, reserve, and regulation markets.

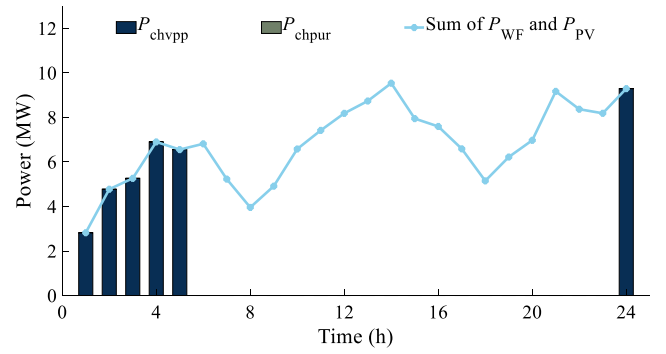


Fig. 11. Charging power of ES and total power of UCPSs.

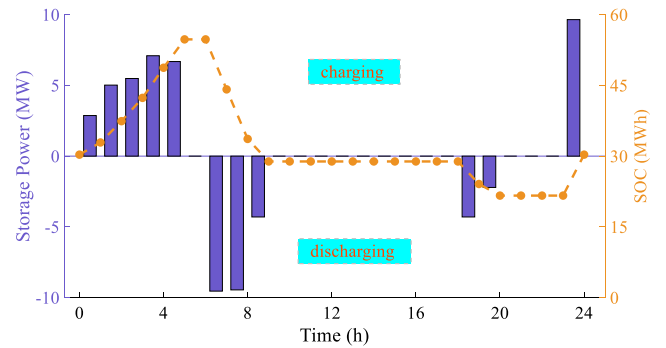


Fig. 12. Charging and discharging power of the ES and the SOC.

significantly higher than that of 2σ MWh. Therefore, in the following bidding strategy and profit analysis, the reserve capacity is selected as 1σ MWh.

4.4. Bidding strategy analysis

In the certain penalty coefficients ($\varphi^+ = 1$ and $\varphi^- = 2$) and the reserve capacity (1σ MWh) of CPSs for UCPSs, the bidding capacities of VPP in each market are solved. It is worth mentioning that the time of problem solving is 0.05 s and the time of modeling through Yalmip is 1.2 s, thus demonstrating its good efficiency for practical application.

Fig. 10 shows the VPP gravitates to participate in the energy market in several suitable hours. The higher-priced reserve market is the most important ancillary market for VPP with no doubt, followed by the regulation down and up markets. It is worth noting that due to the service response of the regulation market is 15 min each time, the bidding capacity seems relatively small. In fact, the VPP dispatches much more generation power for this. The bidding capacities at different time

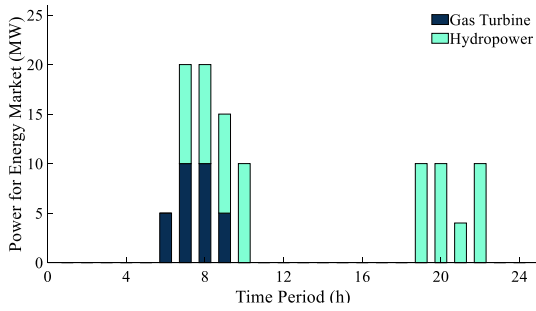


Fig. 13. Scheduling strategy of the GT and HS in the energy market.

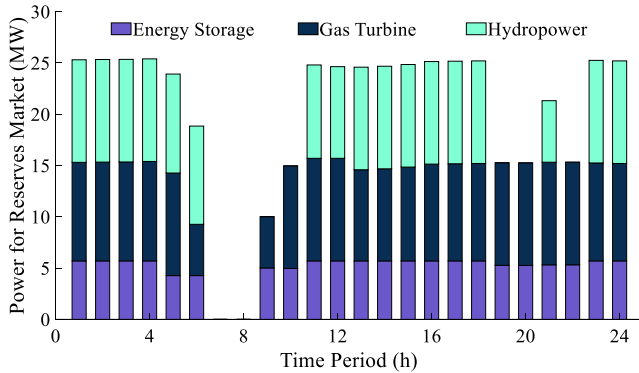


Fig. 14. Scheduling strategy of CPSs in reserve market.

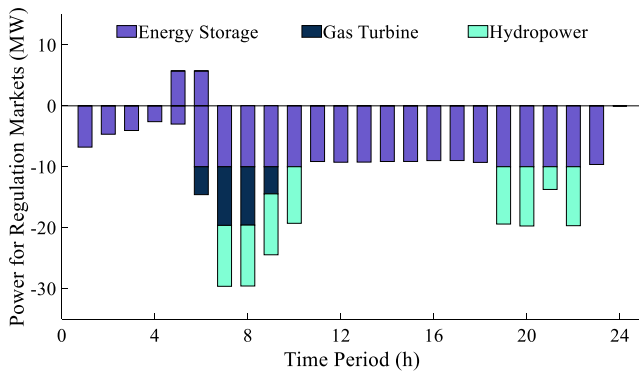


Fig. 15. Scheduling strategy of CPSs in regulation markets.

instants and markets' prices generally show a positive correlation trend. The reason for the capacity allocation of each market can be analyzed through Fig. 11 and Fig. 12.

It is depicted in Fig. 11 and Fig. 12 that the ES will absorb electricity from the UCPSs when the energy price is low (0:00–5:00, 23:00–24:00) and pursue the benefits of VPP by supplying electric power during the higher-priced time period (6:00–09:00, 18:00–20:00). This kind of operation manifests the reasonable utilization of price differences to transfer electricity. Although the energy market price from 4:00 to 5:00 is higher than that from 23:00 to 24:00, the ES will not fully charge from UCPSs in the relative lower-priced period, because of the indispensable reserve down capacity for UCPSs. Besides, the ES will not deeply charge and discharge in one day. On one hand, it is limited by its maximum exchangeable power; on the other hand, it needs to share a part of the capacity to participate in the reserve market for arbitrage. These behaviors are determined by the combined effect of the overall electricity markets, generation power of UCPSs and their charging power, rather than only price difference factor.

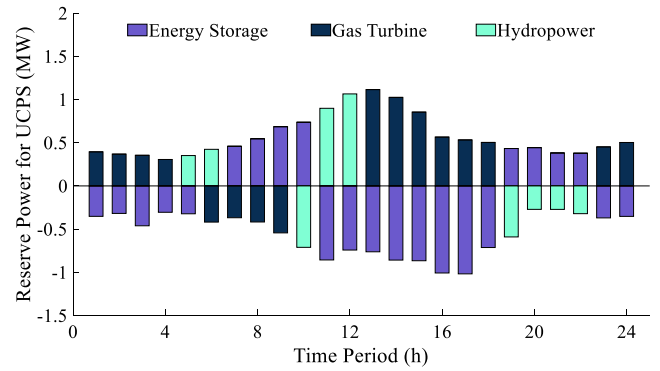


Fig. 16. Reserve capacity of CPSs for UCPSs.

Table 1

Net profit comparison between the single market and multi markets.

Market	Energy only	Energy and Ancillary Service
Net Profit (\$)	42,135	58,925

Later, the behaviors of the HS and GT are illustrated in Fig. 13. The operation time of the HS and GT is relatively short, only in the higher-priced periods, mainly due to their own daily allowed generation capacity and high costs of power generation, respectively. This directly leads to that only ES can participate in the regulation down market and provide the reserve down capacity for UCPSs in remaining time, which embodies the importance of VPP involving with ES. It can be seen from Fig. 14, Fig. 15 and Fig. 16 that the HS and GT can participate in the reserve market, regulation up market and share the reserve up capacity for UCPSs during shutdown hours, which allows the ES not to bear the reserve up capacity for UCPSs from 0:00 to 6:00, and slacks off the upper limit of its discharging output. These interactions evidently reflect the complementary effects among the CPSs.

From 6:00 to 8:00, in addition to providing the necessary reserve capacity for UCPSs, CPSs will participate in the energy market with the maximum output rather than reserve market and regulation up market, due to the higher price of energy market. The ES will participate in the regulation market from 4:00 to 6:00. While due to the power and capacity limitations of constraints in (31) and (37), the optimal result suggests to participate in both the regulation up market and the reserve market concurrently. In the rest hours, VPP prefers the higher-priced reserve market rather than the regulation up market.

4.5. Analysis of VPP's motivation

In addition to the analysis of VPP's bidding strategy, the motivation of VPP should also be investigated to better grasp the changing benefit of VPP. Motivation analysis can explain the reason that VPP should participate in multiple markets, not just the energy market, and why various power sources are willing to form a VPP alliance to participate in the electricity market as a whole. Considering the UCPSs' prediction error, the power generation of the HS, and the retail price of ES to purchase electricity from the grid vary in different quarters, the four bidding days contain more variable parameter information. In the following three quarters of 2019, the last day of each quarter is the bidding day, and the total economic benefits of the four bidding days are

Table 2

Economy comparison with (W)/without (W/O) coordination.

Coordination	Sum of Cost (\$)	Sum of Net Profit (\$)
W/O	32254.0	56989.0
W	26703.0	58898.0

Table 3

Economic comparison under the two behaviors.

Coordination	1st behavior	2nd behavior
W/O	58457.0 \$	59117.5 \$
W	59283.0 \$	60241.1 \$

Table 4

Different combinations of VPPs.

Combination Number	WF	PV	ES	GT	HS
1	✓	✓	–	–	–
2	✓	✓	✓	–	–
3	✓	✓	–	✓	–
4	✓	✓	–	–	✓
5	✓	✓	✓	✓	–
6	✓	✓	✓	–	✓
7	✓	✓	–	✓	✓
8	✓	✓	✓	✓	✓

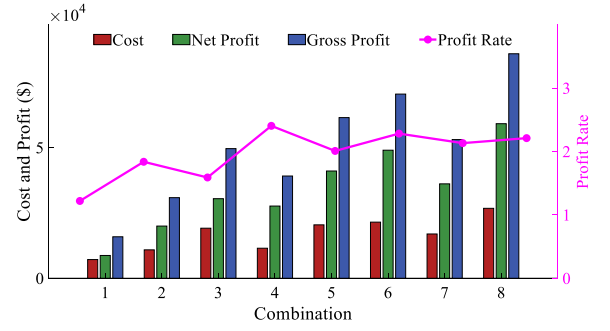
used for economic analysis.

The Table 1 gives an economic benefit comparison of the VPP participating only in energy market and in multiple markets.

From the Table 1, the net profit of VPP participating in multiple markets is obviously higher than only in the energy market, with an increased rate of nearly 40%, which clearly indicates the significance of bidding in multiple markets. In addition, the Table 2 compares the different economies when power sources participate in the markets independently (without coordination) or together (with coordination).

Table 2 shows the joint coordination operation not only reduces the cost by 17.3% but also increases the profit by 3.3%. Through the analysis of the experimental data, the main reason for the significant cost reduction lies in that the ES absorbs a lot of charging power from UCPSS (equals to 4279.1 \$), and CPSs reserve the capacity for UCPSS to reduce the economic penalty cost (1655.2 \$). Furthermore, the reasons for the net profit increase in an alliance can be analyzed from the main behavior of the alliance. There are two typical behaviors: CPSs provide reserve capacity (i.e. the 1st behavior) and the ES transfers the generated electricity of UCPSS into the peak price hours (i.e. the 2nd behavior). Thus, a comparative experiment can be conducted from the following two aspects to quantify their essential contribution to the net profit growth of such alliance. For exploring the 1st behavior, the retail prices of the ES to buy electricity from the grid are set equal to the settlement price of the VPP. Under this situation, the price difference of the ES electricity transferring from UCPSS equals to the sum of the electricity selling profit of UCPSS without the electricity transferring and the price difference of the ES's electricity transferring from the network. Thus, the ES is incapable of bringing economic benefits by electricity transferring. This measure guarantees that the profit growth in the alliance is only affected by the reserve capacity (the 1st behavior). Similarly, for the 2nd behavior, the UCPSS's power prediction is assumed to be absolutely accurate without any deviations, and hence there is no need to reserve capacity to reduce the economic penalty. This ensures that net profit growth under the alliance is only affected by the electricity transferring of UCPSS by the ES. The Table 3 displays the profits under the two behaviors.

As indicated in Table 3, under the influence of two factors separately, the profits of the VPP increase by 1.4% and 1.9%, which shows the significance of reserving capacity for UCPSS by the CPSs and transferring electricity of UCPSS by the ES. In particular, when more reserve capacity is required (if the prediction error of UCPSS becomes relatively larger or the installed capacities of UCPSS are relatively larger compared with the installed capacity of CPSs), or the ES transfers more electricity from UCPSS, the profit growth with coordination will further become more significant.

**Fig. 17.** Economy comparison of different VPPs' combinations.

4.6. Economic comparison on different combinations of VPPs

In reality, not every VPP contains the above five types of power sources. Regarding the WF and PV as the main body, and various CPSs as some additional power sources, a set of comparative experiments for the VPPs composing of different types of additional power sources with the same capacity are studied. The different combinations and economic information are listed in the Table 4 and Fig. 17.

As can be seen from the Fig. 17, when one CPS is equipped for UCPSS, the combination with the highest profit rate (i.e. gross profit divided by net profit) is the VPP composed of the WF, PV, and HS (Combination 4). The reason for that is the HS does not introduce charging costs like ES, and with a lower generating cost than the GT. However, compared to the GT in the same installed capacity of 10 MW, the HS has a lower net profit mainly due to the large one-time investment. When it comes to the ES and GT, although the profit rate of ES is slightly higher, the net profit of ES is less which can be seen from Combinations 2 and 3. When equipping two types of CPS to UCPSS, the ancillary system composed of the HS and GT has the highest profit rate and net profit (Combination 6). In addition, Combination 8 involving with all power sources also performs well in terms of profitability.

Therefore, when it comes to equipping a kind of CPS to UCPSS under the same installed capacity, the introduction of the HS will greatly increase the profits and profit margins of VPP. If hydropower is unavailable due to geographical factors, it is better to select GT from the perspective of absolute profit, but in regard of environmental protection and profit rate, ES will have a higher priority than GT. Importantly, gathering more power sources into the VPP can make more contribution to the economic benefits.

5. Conclusion

This paper studies an optimal bidding strategy model of VPP in context of energy, reserve, and regulation markets in the day ahead, and conducts an in-depth analysis on its operational strategy and economy. The models include UCPSSs such as the WFs and PVs with uncertainty, as well as CPSs like the HS, ES, and GT. This study considers the economic penalty arisen from the prediction power error of UCPSSs, and the countermeasures to reduce the economic penalty through the provision of reserve capacity by these involved CPSs. The effectiveness of the proposed countermeasures is verified via the case studies. Based on the constructed optimization model, the economics of VPPs under different scenarios are performed and discussed through the comparative experiments.

From the theoretical models and the results in the case study, the following conclusions can be drawn:

- (1) Based on the optimal bidding strategy model of the VPP with various power sources participating in multiple markets, this paper analyzes the reasons that the optimal economy can be

achieved through an internal coordination amongst different physical entities inside of VPP. The comparative experiments illustrated the necessity of VPP for aggregating various power sources, especially to participate in the multiple markets.

- (2) Utilizing the quantile variable to perform the reserve capacity from CPS for UCPS and the superquantile theory to quantify the penalty cost due to uncertainty of UCPS, the economic penalty can be reduced by a certain reserve capacity, whereas the overall economy of VPP can be increased. Further, the numerical sensitivity analysis is conducted to effectively identify the optimal reserve capacity condition due to complicated integral calculations in the form of quantiles and superquantiles. The results indicate that the proper reserve capacity can bring remarkable profit improvement about 1.4% in the case of this study.
- (3) From a perspective of the economics, when a kind of CPS is combined with the UCPSs into a renewable-based VPP, the profit rate of the HS behaves better than the ES and GT due to a relatively low cost and high net profit. In the absence of available hydropower, the ES is able to achieve a higher profit rate whereas the GT can bring more net profit.

Based on the works accomplished herein, the future research extension can concentrate how to solve the continuous variable optimization of the reserve capacity.

CRediT authorship contribution statement

Chen Yang: Conceptualization, Methodology, Investigation, Software, Writing – original draft, Formal analysis. **Xiao Du:** Methodology, Investigation, Writing – review & editing. **Dan Xu:** Methodology, Writing – review & editing. **Junjie Tang:** Conceptualization, Methodology, Funding acquisition, Writing – review & editing. **Xingyu Lin:** Conceptualization, Methodology, Writing – review & editing. **Kaigui Xie:** Funding acquisition, Writing – review & editing. **Wenyuan Li:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research work is majorly supported by the National Natural Science Foundation of China (Project No.: 52177071); additional funding is received from the National Science Fund for Distinguished Young Scholars of China (Project No.: 51725701). The authors gratefully acknowledge these financial supports.

Note

Color should be used for all figures in print.

Appendix

The parameters of all the power sources are clearly listed as follows (see Table A1):

Table A1

Parameters of all the power sources.

Power Source (Reference)	Parameter	Value
Wind ([20]) Photovoltaic ([21]) Energy Storage [22–24]	P_{WFmax}	10 MW
	C_{WF}	9.23 \$/MWh
	P_{PVmax}	10 MW
	C_{PV}	10.77 \$/MWh
	P_{ESmax}	10 MW
	S_{bat}	60 MWh
	$C_{bat,ins}$	7.6×10^4 \$
	C_{ESpom}	1×10^4 \$/MW
	r	6%
	T_{bat}	20 years
	C_{EScom}	10 \$/MWh
	κ	[1.85, 1.26, 1.07, 1.63]
	η	0.9
	E_{ESmax}	$0.95S_{bat}$
Gas Turbine [25–26]	E_{ESmin}	$0.05S_{bat}$
	P_{GTmax}	10 MW
	P_{GTmin}	1 MW
	$P_{GT,ramp}$	5 MW/h
	C_{GT0}	25.3 \$
	C_{GT1}	18.15 \$/MWh
	C_{GT2}	0.0087 \$/(MW ² h)
	$\lambda^{su}, \lambda^{sd}$	37.8 \$
	P_{Hsmax}	10 MW
	C_{HSom}	4.5 \$/MW
Hydropower [27]	$C_{Hsd,ins}$	1.641×10^6 \$/MW
	T_{HS}	40 years
	Q_{Hsmax}	[74, 76.5, 57.2, 35.6] h

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